

ON THE THEORY OF FERROMAGNETIC RESONANCE

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A ferromagnetic material, magnetized up to its saturation value M_0 by means of a constant external field H_0 , shows interesting resonance phenomena when small additional magnetic fields of high frequency are applied. Kittel¹⁾ has been able to give an explanation of the effect by using classical equations of motion for the magnetic dipoles. In the special case of a non-conducting ellipsoidal specimen, magnetized in the direction of one of its principal axes (z -axis) it is easy to derive an expression for the induced alternating magnetization m when a small external alternating field $h_x \exp(j\omega t)$ is applied, provided that the dimensions of the ellipsoid are small in comparison with the wavelengths of the electromagnetic waves inside the material. Neglecting damping terms and anisotropy effects one finds²⁾ (exponential time factors being omitted):

$$[\gamma^2 \{H_0 + (N_x - N_z) M_0\} \{H_0 + (N_y - N_z) M_0\} - \omega^2] m_x = \\ = \gamma^2 M_0 \{H_0 + (N_y - N_z) M_0\} h_x \quad (1)$$

$$[\gamma^2 \{H_0 + (N_x - N_z) M_0\} \{H_0 + (N_y - N_z) M_0\} - \omega^2] m_y = j\omega\gamma M_0 h_x$$

where N_x, N_y are the demagnetizing factors in the x, y directions and so on, and where γ should be equal to the gyromagnetic ratio of the spins. This result gives the dependence of the resonance frequency on the shape of the specimen.

We have found that a phenomenological description of ferromagnetic resonance can be given which does not involve explicitly the shape of the body. We consider the parts of the vectors B and H which are periodic in time. These parts will be called B' and H' . They satisfy Maxwell's equations inside and outside the material and the usual boundary conditions are valid. If z is the direction of permanent magnetization, the relation between B' and H' inside the ferro-

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magnetic material can be written as (exponential time factors being omitted):

$$\begin{aligned} B'_x &= \mu H'_x - jaH'_y \\ B'_y &= \mu H'_y + jaH'_x \\ B'_z &= H'_z \end{aligned} \quad (2)$$

The quantities μ and a can be calculated from the same classical model as used by Kittel. We find

$$\mu = \frac{\gamma^2 H_i B_i - \omega^2}{\gamma^2 H_i^2 - \omega^2} \quad a = \frac{4\pi M_0 \gamma \omega}{\gamma^2 H_i^2 - \omega^2} \quad (3)$$

where H_i and B_i are the (constant) magnetic field strength and magnetic induction inside the material due to the constant magnetizing field.

With the aid of these relations all results obtained by Kittel²⁾ for small ferromagnetic ellipsoids can be derived again. A further application is the calculation of the index of refraction of an electromagnetic wave propagating through an (infinitely large) specimen of non-conducting magnetized ferromagnetic material. We have found that for every direction of propagation and for every frequency there exist two different, elliptically polarised waves with different indices of refraction. The refractive index of one of these two waves shows a resonance phenomenon at a frequency given by

$$\omega^2 = \gamma^2 H_i (H_i + 4\pi M_0 \sin^2 \vartheta);$$

ϑ is the angle between the direction of propagation and the z -axis. The other waves do not show resonance. In the special case $\vartheta = 0$ there are two circularly polarised waves, and in the case $\vartheta = \frac{1}{2}\pi$ there are two linearly polarised waves.

In order to justify the use of classical equations of motion, we have derived the equations (1) from quantum mechanical considerations. We use as a starting point a Hamiltonian containing terms describing the interaction of the spins with the external magnetic fields, an isotropic exchange interaction between nearest neighbours and the magnetic dipole-dipole interaction between the spins in the small ellipsoid. When the constant external field is sufficiently high and the temperature is sufficiently low only those stationary states of Schrödinger's equation will play a role for which there is nearly complete alignment of the spins. Holstein and Primakoff³⁾ have developed an approximate method for dealing with

these states and have determined the corresponding eigenvalues of the energy in a first approximation. Their method can be considered as an improved Bloch spin-wave treatment of ferromagnetism; the improvement consists in including the dipole-dipole terms.

When we now consider the additional alternating field as a small perturbation, we can determine m_x and m_y by means of a method analogous to that used in the theory of coherent scattering. Resonance will occur at frequencies for which $\hbar\omega$ is equal to the difference of the energies of two stationary states between which the alternating magnetic field can cause transitions. We find that in H and P's approximation the energy difference between any two of these states is always equal to $2\beta [\{H_0 + (N_x - N_z) M_0\} \{H_0 + (N_y - N_z) M_0\}]^{\frac{1}{2}}$ and that the approximation eventually leads to expressions for m_x and m_y which are identical with the classical expressions if we put $\gamma = 2\beta/\hbar$. Here β is the magnetic moment of the spins and thus γ is equal to the gyromagnetic ratio in the case of free spins. By taking into account terms in the Hamiltonian, which are neglected in H. and P.'s approximation it should also be possible to estimate the width of the resonance line.

Our quantum mechanical consideration suggests a possible explanation for the observed discrepancy ⁴⁾ between the value of γ determined by resonance experiments and the gyromagnetic ratio. When we assume that the combined effect of a crystal electric field and spin-orbit interaction causes a deviation of the effective magnetic moment β of the spins from the spin only value μ_B , let us say $\beta > \mu_B$, then $\gamma > 2\mu_B/\hbar$. At the same time the effect will cause a still larger deviation of the effective angular momentum of the spins from the spin only value $\frac{1}{2}\hbar$. Therefore the ratio of the effective magnetic moment and the effective angular momentum (gyromagnetic ratio) may be expected to be smaller than $2\mu_B/\hbar$. It would be interesting if this connection between the deviations of the angular momentum and γ from spin-only values could be verified experimentally. A fuller account of our calculations will be published in the *Philosophical Magazine*.

REFERENCES

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